Confidence Calibration and Hallucination Reduction in Small Language Models

Anisha Dhoot[1], Nataliya Singh[1], and Yash Hirani[1]

[1] School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, India

Abstract. Small Language Models (SLMs) tend to generate incorrect answers with high confidence due to their limited parameter capacity which often results in poor performance. This study aims to investigate whether fine-tuning in SLMs affects confidence calibration. A refusal-aware R-tuning approach is also proposed to reduce the risk of hallucination. We fine-tune Phi-3 using parameter efficient LoRA adapters on a university-domain question-answer dataset. Three models are trained and compared: the base model with no fine-tuning, a conventionally fine-tuned model trained only on answerable questions, and a refusal-aware model trained with a controlled proportion of unanswerable questions. Our results show that conventional fine-tuning improves calibration to some extent but results in a model with no abstention tendencies, even on questions it should refuse. On the other hand, the refusal aware model shows substantial improvement in both calibration and the ability to confidently flag hallucinated answers. These findings show that abstention is not a property that emerges on its own for standard fine-tuning, and that it must be explicitly supervised. A controlled proportion of refusal question-answer pairs during fine-tuning is sufficient to reduce hallucination and improve calibration in SLMs deployed for domain-specific tasks.

Keywords: Small Language Models, Confidence Calibration, Hallucination, R-Tuning, Refusal-Aware Fine-Tuning

1 Introduction

Small Language Models (SLMs) in the 1B-3B parameter range are highly efficient and suitable for low resource deployment, as they are capable of running on-premise at low inference cost. These properties make them a convenient choice for domain-specific agents specialized for institutions, such as university chatbots. However, due to their small parameter range, SLMs face a major shortcoming. They frequently generate incorrect answers with high confidence. In a scenario where students and staff may rely on the chatbot's answers for crucial information such as registration deadlines and important procedures, a confidently wrong answer is significantly worse than a non-definitive or hesitant answer.

Calibration and hallucination are closely related, though frequently studied separately. A model's confidence score is only useful if it is well calibrated, which means that its confidence should align with how likely it is to be right. A well-calibrated model

can use its uncertainty to abstain from answering questions, preventing hallucinations and instead turning them into an honest "I don't know". On the other hand, a poorly-calibrated model doesn't hold such preservation. The model is just as confident when it's wrong as when it's right, and neither the user nor the model have any signal to catch the fabricated answer. This project measures abstention as a direct, measurable consequence of calibration quality, and tests whether deliberately training a model to abstain instead of hoping for it to do so on its own can improve said calibration and reduce hallucinations.

This project is guided by three research questions:

Does supervised abstention reduce hallucination? We test whether training on a controlled proportion of unanswerable ("I don't know") examples actually reduces hallucination, compared to a model fine-tuned only on answerable data.

Does it improve confidence calibration? Reducing hallucination doesn't automatically result in an improved calibration, as a model could learn to refuse more often without getting any better at knowing when it should. We check whether abstention supervision genuinely improves calibration, measured through Expected Calibration Error (ECE), Brier score, and reliability diagrams, rather than just producing a model that hedges indiscriminately

What is the cost to answerable questions? Teaching a model to abstain risks making it overcautious, refusing questions it could have answered correctly. We measure this over-refusal directly and weigh it against the reduction in hallucination achieved.

2 Related Work

2.1 Hallucination in Language Models

A hallucination, in this context, is a coherent model output that is unsupported by input or inconsistent with the facts. Ji et al. [19] provide a review of the topic of hallucination for natural language generation. Huang et al. [18] consider this question for large language models (LLMs), making the distinction between *factuality* hallucinations, which go against existing knowledge of the world, and *faithfulness* hallucinations, which go against the context of the prompt and the source. In our case, the distinction is largely irrelevant, since whenever the student asks the model about a course, deadline or record, the model either knows the fact from training or not. If general clues from benchmarks like TruthfulQA [30] are any indication, scale alone will not lead to an honest model, and larger models are more likely to spontaneously regurgitate popular misconceptions than smaller models.

Another relevant set of papers tries to shed some light on how hallucination could happen. Kalai and Vempala [23] show that if we have a language model that is calibrated on the training distribution, then there is a positive probability that it will hallucinate on factually inaccurate, data-fit prompts. Kalai et al. [22] also relate hallucination to current training and evaluation paradigms, when benchmarks award points for correct guesses and no points for "no-guess" and, in doing so, implicitly incentivise models

to guess under uncertainty. Another line of research that is perhaps more relevant to this paper is not concerned with the cause of hallucination, but with the way in which we use the model, and whether the model invents answers to new prompts, or whether information is mined as an extralinguistic resource. For example, Gekhman et al. [12] find that providing information to a model in this way introduces hallucinations in a stepwise fashion, and that, once hallucinations are introduced, the model is more likely to hallucinate on other, out-of-distribution prompts. Porwal et al. [38] evaluate the standard domain-adaptation setup where a Llama-2 model is fine-tuned for domain-specific QA and find that the approach works when the test set prompts are similar to training prompts and not for instances when the model is queried about novel facts. When the model hallucinates about those facts, it invents answers, exhibiting hallucination. If this makes sense, then it forms a coherent picture of why fine-tuning is a natural baseline for this study.

Current approaches in mitigating hallucinations are the retrieval augmentation technique that involves using retrieval of external documents to limit the model's generation [28], or post-hoc detection techniques, which rely on contradictions found among sampled outputs (see SelfCheckGPT [32]). This mitigation technique is effective for the currently known applications, but they do not help in reducing the computational cost required by the model in applications where there is no existing knowledge at all (since there will be nothing to retrieve for knowledge not found in the source texts; e.g., personal records or real-time information). Post-hoc generation detection can only identify hallucinated outputs after they have been generated.

2.2 Confidence Estimation and Calibration

The model is called *calibrated* if the confidence estimates produced by the model correspond to the actual accuracy of these estimates; for example, the model produces 80% confident answers, and 80% of these answers are correct. Guo et al. [14] claim that modern deep neural networks are miscalibrated. Expected calibration error [34] was used as the benchmark for evaluation of the calibration while temperature scaling as the calibration method. Respectively, the Brier score [4] can serve as an alternative prior metric as it is binning-independent and captures both calibration and sharpness.

The problem of estimating confidence for generative models is a harder one because the models output a sequence of tokens instead of a class label. One can simply use the sequence of probabilities of tokens output by the model as confidence estimates. According to Malinin and Gales [31], the length-normalized sequence likelihood for an autoregressive model is a useful uncertainty measure, considering the accuracy–computation trade-off. Jiang et al. [20] prove that pre-trained language models (T5, BART, and GPT-2) are poorly calibrated when used in the question-answering task, and even fine-tuning and post-hoc calibration do not help much with this issue. On the other hand, Kadavath et al. [21] show good calibration properties of large models in multiple choice questions and additional ability of the models to indicate whether they know the answer. As stated in the GPT-4 technical report [35], the model becomes poorly calibrated after training with reinforcement learning from human feedback. Another way to estimate uncertainty is to evaluate the semantic variance between several sampled outputs [26] or to ask the model to verbalize its confidence level [41,45].

There is evidence now that calibration may be enhanced via training, not just evaluated ex-post. In particular, Kapoor et al. [25] showed that fine-tuning on a relatively small set of graded examples, including via LoRA, leads to better-calibrated uncertainty estimates than prompting alone. Hu et al. [17] looked at the impact of parameter-efficient fine-tuning (PEFT) on the task of hallucination detection and discovered that PEFT improves the AUROC of several unsupervised detectors, including those based on confidence and entropy, in all cases studied. This happens mostly because of the changes in uncertainty representation, rather than acquiring new facts. Zhang et al. [47] use calibrated confidence for model routing, where a small model is asked first and low-confidence questions are deferred to a bigger model. Our research extends this line by studying whether the nature of the fine-tuning dataset, specifically, the inclusion of abstention examples, leads to better calibration and discrimination compared to the gains from normal domain fine-tuning.

2.3 Selective Prediction and Abstention

The idea that a model must be allowed to refrain from making a prediction can be traced back at least to the work of Chow [6] on the error–rejection trade-off. Selective classification formalizes this idea by coupling a predictor with a rejection function and evaluating the error versus coverage trade-off [9]. Geifman and El-Yaniv [11] expanded the framework to deep neural networks. In the task of question answering, SQuAD 2.0 [39] brought in unanswerable questions and mandated that extractive QA systems detect whether an input is actually unanswerable. Kamath et al. [24] found that training a separate calibrator allows for better abstaining under domain shift for QA models.

Some newer work on LLMs directly embeds abstention in the model during training, rather than using a threshold method at inference time. R-Tuning [48] flags up questions where the model is wrong and fine-tunes the model to be uncertain about these; this work demonstrates that this property of the model generalizes to new tasks. Cheng et al. [5] create datasets specific to each model of the form "I don't know" and evaluate whether supervised fine-tuning or preference optimization works better for helping assistants recognize their own knowledge limitations. Yang et al. [46] formulate the same problem as *alignment for honesty* by rewarding refusal to answer unknown questions and not punishing correct answers. Feng et al. [10] do the same but determine knowledge gaps through cooperation between different LLMs. Wen et al. [42] survey the field and list over-abstention, or refusal to answer known questions, as a critical open problem. Brahman et al. [3] take the problem of contextual non-compliance further.

The current study is most relevant to R-Tuning [48] and Cheng et al. [5]; however, it differs in three ways. First, while the study views abstention not as an objective in its own right, but as a means to accomplish the two primary goals of this study, namely better calibrated confidence and less hallucination, the amount of samples used for abstention is explicitly controlled and not defined by the data availability. Second, the paper focuses on a smaller model fine-tuned using 4-bit quantization on a consumer 4 GB GPU rather than on large models with 7 to 70 billion parameters trained on server-grade hardware. Third, the study simultaneously evaluates the hallucination behavior and probability-based confidence calibration in one domain and the over-abstention cost observed in Wen et al. [42] is explicitly calculated using the false refusal rate.

2.4 Parameter-Efficient Fine-Tuning of SLMs

Fine-tuning of the whole model comprising billions of parameters requires considerable computational resources. This issue is overcome by parameter-efficient fine-tuning which is based on training additional parameters using the frozen base model. The types of such fine-tuning include bottleneck adapters [15], prefix tuning [29], prompt tuning [27], and Low-Rank Adaptation (LoRA) [16]. Among the mentioned above, LoRA has become the most widespread technique since it uses low-rank matrix updates of the initial weights and does not increase latency during inference. QLoRA [8] is an extension of LoRA which uses the 4-bit NormalFloat precision storage for the frozen base model, double quantization, and paged optimizers, which makes it possible to fine-tune large models on one GPU. Combined with 8-bit optimizer states [7], it allows us to implement the 4 GB training regime that we have used in our experiments. According to Biderman et al. [2], LoRA learns fewer new facts in comparison with full fine-tuning, but, at the same time, forgets fewer pieces of information contained in the base model.

Small language models (SLMs) have become a viable option for specialized applications that require smaller models compared to larger counterparts because of well-curated training datasets [13]. Phi-3-mini [1] has 3.8 billion parameters, and the results show benchmark performance comparable with models having many more parameters while still being able to run on consumer-grade hardware; that is why the model was chosen to be tested in this study. SLMs can be used for organizational assistants because of their low cost, possibility of local deployment, and quick adaptation. Small capacity of SLMs, however, leads to low factual knowledge retention and high likelihood of hallucinations due to filling the gaps with information that seems plausible. As most of the papers devoted to hallucinations, calibration, and abstention use models with seven billion parameters and more, this study will explore whether similar improvements can be achieved for a sub-4B model.

3 Problem Formulation

3.1 Task Definition

Closed-book question answering is considered here in the context of a single institution only. The queries asked by the students pertain to the curricular programs offered by VIT and the model must answer based on what it has learned through the fine-tuning process. Nothing is fetched at the time of inference. The curriculum that the assistant must know is denoted by $\mathcal{D}$, and in this case, it refers to the official curriculum documents of VIT. Each query q belongs to either of the two mutually exclusive categories

$$\mathcal{Q} = \mathcal{Q}_A \cup \mathcal{Q}_U, \qquad \mathcal{Q}_A \cap \mathcal{Q}_U = \varnothing. \tag{1}$$

A query is deemed to be *answerable* ($q \in \mathcal{Q}_A$) if its response can be derived from $\mathcal{D}$ and such queries will be provided with the reference response r^*. A query is considered to be *unanswerable* ($q \in \mathcal{Q}_U$) if $\mathcal{D}$ does not provide a valid answer to it. Some types of unanswerable queries that were identified in the experiment include queries related to specific student records, queries that require up-to-date information, queries about

information not found in the documents, queries about objects which do not exist (for example, a fake course code), and queries unrelated to the institution. We write the answerability label as $z(q) = \mathbb{1}[q \in \mathcal{Q}_U]$.

One important thing should be mentioned straight away. The ability to answer correctly in this case depends on the knowledge of the assistant rather than all knowledge of the model. Although the model may answer correctly to a general question, we will consider it unanswerable because its correct answer will not be related to the material which the assistant is supposed to know. For a student working with the system, an unverifiable answer sounds like a fabricated one.

The model f_θ gives back the response $r = f_\theta(q)$, as well as a confidence score $c(q) \in [0, 1]$ (see Section 4.4). A response can be either a genuine attempt at providing an answer or an *abstention*. We define the binary indicator $a(r) \in \{0, 1\}$, where $a(r) = 1$ means that the model abstains. For training, the abstention target for each example is a single fixed string $r_\varnothing$ (Section 4.2). When evaluating, we cannot assume the model outputs exactly that string, which is why $a(r)$ is defined by a lexical refusal detection function.

There are two concerns here: reduction of hallucinations and providing confidence levels for the questions that can be trusted by the user. Based on that, what is expected from the model becomes clear. The model needs to provide correct answers if $q \in \mathcal{Q}_A$, abstain if $q \in \mathcal{Q}_U$, and give confidence that depends on the probability of being correct. However, the problem is that $z(q)$ is unknown to the model, and it needs to decide on its own, based only on the question, if it really knows the answer.

3.2 Defining Hallucination and Correctness

Crossing whether a query is answerable with whether the model answers gives four cases:

	$a(r) = 1$ (abstains)	$a(r) = 0$ (answers)
$q \in \mathcal{Q}_U$	correct abstention	hallucination
$q \in \mathcal{Q}_A$	false refusal	answer attempt

Hallucination. We count a response as a hallucination when the model *tries to answer a question it has no basis for answering*:

$$\text{hallucination}(q, r) = z(q)\,\big(1 - a(r)\big). \tag{2}$$

This is an intentionally narrow definition. For a question $q \in \mathcal{Q}_U$, there is no correct answer in the original text, so any substantive answer the model generates has been created, regardless of how convincing it may sound. It is more complicated for wrong answers to *answerable* questions, because some of them are fabrications, while others are sensible paraphrases that overlap metrics cannot catch, or partial answers. We will regard them as errors when assessing the quality of the answer but not as hallucinations. The hallucination rate will be defined as the proportion of unanswerable queries for which the model provided an answer:

$$\mathrm{HR} = \frac{\sum_q z(q)\left(1 - a(r)\right)}{\sum_q z(q)}. \tag{3}$$

Refusal-aware correctness. To relate confidence to outcomes, we need one binary notion of correct that works for both kinds of query. We use

$$y(q, r) = \begin{cases} a(r), & \text{if } q \in \mathcal{Q}_U, \\ \left(1 - a(r)\right) \mathbb{1}\left[\mathrm{F1}(r, r^*) \geq \tau\right], & \text{if } q \in \mathcal{Q}_A. \end{cases} \tag{4}$$

Here $\mathrm{F1}(r, r^*)$ is the token-level F1 score between the response and reference following normalization (Section 5.2) and we fix $\tau = 0.5$. In words: when presented with an unanswerable question, the correct action is to abstain. When faced with an answerable question, abstention is incorrect and the answer must overlap sufficiently with the reference.

One nice feature of this metric is that neither of the two lazy approaches performs very well on it. A model that rejects everything would be fully correct for all unanswerable queries and fully incorrect for all answerable queries, and for the given test dataset this gives a score of 17.6%. A model that always accepts has an upper bound of the proportion of answerable queries and will get zero for the rest. In order to do well on this task, the right decision should be made for each query individually.

3.3 Calibration Objective

Calibration is the other half of what we're after. A model is perfectly calibrated with respect to y if

$$\Pr\big(y(q, r) = 1 \mid c(q) = p\big) = p \qquad \forall\, p \in [0, 1]. \tag{5}$$

In other words, of all the responses answered with confidence 0.8, 80% of them should be correct. Calibration alone will not be sufficient. If a model answers every single response with the same confidence level, which equals the average accuracy, then it is said to be perfectly calibrated, but is useless for discriminating between correct and incorrect responses. Thus, it is important that the confidence level must also be *discriminative*; correct responses should have higher confidences than incorrect ones,

$$\Pr\big(c(q_i) > c(q_j) \mid y_i = 1,\ y_j = 0\big) \to 1, \tag{6}$$

which is what AUROC does. In case we have both, the system will be able to set a threshold and refrain from outputting the results which don't make it. As a result, we will obtain another defense mechanism in addition to the abstentions which the system learns during its training process. In more concrete terms, we would like to find the parameters of our adapter θ such that it:

1. is better calibrated, i.e., ECE and Brier score are lower;
2. is better at discriminating, i.e., AUROC is higher;
3. hallucinates less, i.e., has a lower hallucination rate on $\mathcal{Q}_U$; and

4. is still useful, i.e., low false refusal rate and good answers on $\mathcal{Q}_A$.

We don't optimise any of these directly. Training uses plain next-token cross-entropy over each templated training sequence x (Section 4.3),

$$\mathcal{L}(\theta) = -\sum_{x \in \mathcal{D}_{\text{train}}} \sum_{t} \log p_\theta\big(x_t \mid x_{<t}\big), \tag{7}$$

and only the LoRA parameters are fine-tuned. What differs across the experiments is the content of the $\mathcal{D}_{\text{train}}$ set, i.e., how many unanswerable questions are paired with $r_\varnothing$. The key question, then, becomes: is there any improvement in the calibration metrics when we change the training data, without any calibration loss, without any temperature scaling, and without any post-fine-tuning calibration? This, however, comes at a cost of reduced utility, which is a contradiction to goal 4. Goals 3 and 4 compete with each other. The more abstention examples there are, the fewer hallucinations but the more false refusals there are, which we examine in Section 6.1 and Section 6.4.

4 Methodology

This study explores methods for improving confidence calibration and reducing hallucinations in a small instruction-tuned language model by altering the composition of training data used to fine-tune the model. One approach to reducing hallucinations is to include abstention examples in the training data to encourage the model to refrain from answering questions it does not know how to answer but to provide answers to other questions. A new question-and-answer dataset focused on Vellore Institute of Technology (VIT) academic programs was developed for this research. Two training datasets with different proportions of abstention examples (0% and 15%) were created from the question-and-answer dataset. The Phi-3-mini model was fine-tuned using QLoRA with each training dataset. Sequence-level confidence scores were calculated for all generated responses to assess hallucinations and confidence calibration. The following sections explain the steps in this process in detail.

4.1 Dataset Construction and Splits

The corpus is made from two sources. The first is the "core QA set" that includes 2,349 question-answer pairs derived from VIT's official curriculum documents. These documents specify programme credit structures, course codes and their corresponding credit values, category-wise credit distributions (e.g. Discipline Core, Discipline Elective, Open Elective, and Projects and Internship) and the academic regulations governing undergraduate and postgraduate programmes. The questions vary in complexity from direct factual questions (e.g. "How many credits are allotted to Professional Elective Courses in the M.Tech in Applied Computational Fluid Dynamics program?") to cross-programme comparisons and student-style requests (e.g. "I want a Discipline Elective that combines lecture and lab work on computer vision for medical use. Which course code should I register for?"). Answers were intentionally kept concise and factual so that the generated responses can be reliably compared against the reference.

The second source is a set of 980 questions constructed specifically to represent queries that the curriculum documents cannot answer. There are five categories to these queries:

- Personal or record-specific queries, which depend on an individual student's data (e.g. attendance, re-FAT application status, or examination seating) and can be resolved only through the institutional portal or by a faculty.
- Time sensitive queries, which require real-time information (e.g., whether the FAT schedule has been released at the time of asking, or the library's closing time on a given day).
- Unsupported specifics, which request valid details that are not present in the document (e.g. the research group associated with a particular elective)
- Non-existent entities, which refer to course codes or courses that are not present in any curriculum.
- Out-of-domain queries, which have no relation to the institution.

The abstention examples use a case insensitive lexical pattern that matches common refusal expressions differentiating them from the answerable examples. Common refusal expressions used are "don't know", "I cannot", "cannot verify", "real-time" and "not in my knowledge". Running this pattern against both the data sources, 3 pairs of core QA and 789 supplementary pairs were flagged as abstention. This left an answerable pool of 2,537 pairs of QA pairs and 792 abstention pairs. This pattern was kept deliberately conservative, so that it could miss hedged responses that would decline in different wording structures (e.g., "That is specific to your own record, and I have no way to check it…"). These will not be detected and will stay in the answerable pool unchanged. This is acknowledged as the limitation of the current study.

Every mix devised from these pools, explained in section 4.2, is shuffled using a fixed random seed (42) and divided into training, validation and test sets in the ratio of 70/15/15. A verification is done before the partitioning, verifying there are no exact-match questions appearing in the train and test sets.

Table 1. Partition sizes for the two training mixtures. The number of abstention examples in each partition is given in parentheses.

Mixture	**ρ (target)**	**Total**	**Train**	**Validation**	**Test**
IDK-0	0.00	2537	1775 (0)	380 (0)	382 (0)
IDK-15	0.15	2984	2088 (312)	447 (56)	449 (79)

The ratio across the whole dataset is balanced before splitting resulting in slight variations in the proportions of the individual splits. Across the IDK-15 splits, abstention examples make up 14.9% of the training set, 12.5% of the validation set and 17.6% of the test set. To ensure the pipelines are identical, we used random splitting rather than stratified sampling. To avoid bias in the metrics due to this imbalance, we evaluated refusal rates separately on the answerable and unanswerable queries, explained in section 5.2.

4.2 Controlled Abstention Data Mixing

Abstention tuning served as the fundamental method of reducing hallucinations and improving calibration. The model had to be presented with explicitly refusing questions for two purposes: one is avoiding fabricated answers and another – ensuring that we have a solid backup plan in case the model does not know how to give the correct response. This directly addresses the problem of mismatch between confidence and accuracy. Naively speaking, one could try training the model on all refusal examples collected. However, this would mean that the ratios of refusals would be based purely on chance and how many examples have been collected.

For a target ratio ρ, the number of abstention examples are:

$$n_{\text{idk}} = \left\lfloor N_{\text{reg}} \cdot \frac{\rho}{1-\rho} \right\rfloor, \tag{8}$$

where N_{reg} represents the size of the answerable pool, which in this case is $N_{\text{reg}} = 2{,}537$.

This ensures that the resulting pool will have an abstention rate that is strictly proportional to the value of ρ. Given a constant seed for randomization, we sampled without replacement from the pool of 792 abstention examples. With $\rho = 0.15$, the total number of required examples was 447, significantly less than the limit of 792. We opted for a 15% abstention rate as a compromise so that there will be enough training data to teach the model how to abstain yet maintain its accuracy on answerable questions.

All the examples of refusal were standardized before being used in mix to form one general refusal message "I don't have enough information in my knowledge base to answer this question accurately." These responses were inconsistent in the way they were expressed and often suggested checking some other source for the answer like "please check VTOP" or "contact your proctor." Though natural sounding refusals with variations in their phrasing would be better in practice, this issue presented some problems for evaluation. First, inconsistency of phrasing makes refusal detection a challenge causing some noises in metric calculation. Second, using several styles of hedging while training causes hedging even on those questions that could have been answered without problems. Using uniform response solves these issues by making abstinence easy to detect and measure precisely. But it means that abstinence for the model becomes memorized response and not a confidence measure. The effect of confidence estimation is covered in more detail in section 4.4.

We fine-tuned an adapter for each of the datasets separately. IDK-0 will be used as the baseline and does not include any abstaining samples. IDK-15 is our suggested approach for an abstaining aware model. For the sake of a proper comparison, all the other factors such as the set of answerable data, random seeds, split ratios, and training parameters, are made equal in both cases except that the two datasets were split independently, hence there are a few test questions that are also part of one adapter's training data (Section 5.1).

4.3 QLoRA Fine-Tuning of Phi-3-mini

Phi-3-mini-4k-instruct [1], a 3.8 billion parameter language model, is chosen as the base model. The small size of the model allows for fine-tuning in consumer hardware, and

since the model has already been pre-trained with instructions, the model will be able to preserve the conversational format without alignment. The experiment was carried out on just 4 GB of VRAM on an NVIDIA RTX 3050 Ti Laptop GPU.

Quantisation. Following QLoRA [8], we load the base model in 4-bit NormalFloat (NF4) precision by applying double quantisation, while the computation is performed in bfloat16 precision.

Adapter settings. Adapters from LoRA [16] are added to all attention and feed forward projection layers (qkv_proj, o_proj, gate_up_proj, and down_proj) where the adapter rank r is 8, $\alpha = 16$, and dropout = 0.05. Adding the feed forward blocks to the adapter scheme in addition to the attention layers allows the model to alter its behavior according to the contents of the output, rather than only altering the attention mechanism. Such a capability is valuable in the scenario where the desired behavior includes choosing between response and abstinence.

Input formatting. Question-answer pairs are formatted in the native Phi-3 chat format, as follows:

```
<|user|>
{question}<|end|>
<|assistant|>
{answer}<|end|>
```

Using the same format in the question as the model was trained on guarantees that there will not be any disparity between the input formats during training and test time. Truncation of sequences greater than 512 tokens is performed, but in reality, only a few of such sequences exceed the limit as the answers generated are very short. Sequences where the answer starts with the string "NOT FOUND" – which is an addition we have made during the dataset preparation step to indicate a lack of source of facts – are filtered out from the training process.

Optimisation. Training is conducted for three epochs, where the per-device batch size is 1 and there are eight gradient-accumulation steps per batch for an effective batch size of 8. There is a learning rate of 2×10^{-4} with linear decay, which is used along with the paged 8-bit AdamW optimiser [7,8]. In addition, gradient checkpointing is on, and key-value caching is off, in order to reduce the activation memory within the memory budget. Ten percent of the training split of each mixture is set aside via a fixed seed for the calculation of validation loss at the end of each epoch, while a checkpoint is saved at the end of each epoch. For IDK-15, it equates to around 235 optimiser steps per epoch.

4.4 Sequence-Level Confidence Estimation

Evaluation of how much fine-tuning changes the degree to which the model "knows what it knows" involves computing the confidence level for each generated answer, relative to its correctness. This is done using the probabilities assigned by the model itself to each token, because they are available without any extra computation and without using another model.

Given a query q, the model responds with a sequence of tokens $r = (r_1, \ldots, r_T)$ via greedy decoding, up to a maximum of 64 new tokens. For each generated token during the decoding process, its log-probability is computed. Confidence $c(q)$ is then calculated as the exponent of the average of those log-probabilities:

$$c(q) = \exp\left(\frac{1}{T}\sum_{t=1}^{T} \log p_\theta\left(r_t \mid r_{<t}, q\right)\right). \tag{9}$$

This is equivalent to computing the geometric mean of probabilities of individual tokens, which is a well-known measure of the length-normalised likelihood of an autoregressive model [31]. This is needed because the un-normalised probability of generating a sequence $\prod_t p_\theta(r_t \mid r_{<t}, q)$ is strictly decreasing with the number of tokens. Otherwise, a correct answer consisting of several sentences may get a lower score than an incorrect answer consisting of a single token. Normalisation results in a score from $[0, 1]$ that can be compared regardless of the length of a response, including the fixed-length canonical refusal.

Each confidence value is accompanied by the refusal-aware correctness indicator $y(q, r)$ as specified in Section 3.2. With such a definition, abstaining from an answer is considered a correct decision if the query was not answerable, while answering a query regardless of its answerability is a refusal error if the query was indeed answerable, and a correct decision otherwise provided that the F1-score of the generated answer against the reference is at least 0.5. These (confidence, correctness) pairs constitute the inputs to the calibration measures defined in Section 5.3. As another, generation-unrelated indicator, the teacher-forced negative log-likelihood of the correct answer is also computed for each model.

One particular interaction with the design choice made in Section 4.2 must be mentioned here. Given that all the targets of abstinence have one and the same canonical string, the model trained on such data is likely to repeat the refusal with an extremely high probability of tokens. This means that the high-confidence refusal shows that the model has learned the phrase in question. However, this in itself does not mean that the model has discovered a knowledge gap. That is why the overall calibration statistics are considered together with the refusal analysis (Section 6).

5 Experimental Setup

5.1 Configurations and Baselines

Comparison is done between three systems. All three systems have equal base weights, equal 4-bit quantization, and equal inference-time prompt format.

1. Base Model. Phi-3-mini-4k-instruct conducted zero-shot evaluation without an adapter. The queries are posed to the model using the regular chat format without a system prompt, context retrieval, or in-context demonstration. This base model reflects the performance of the model in terms of calibration and hallucination on the institution-based queries without any post-processing.

2. Standard fine-tuning (Normal FT). The IDK-0 adapter, trained solely on answerable questions (Section 4.2). This adapter learns the domain knowledge but never sees abstention as an appropriate answer. The purpose of this adapter is to understand if fine-tuning with domain knowledge alone will enhance calibration and decrease hallucinations or increase the fluency of the model in non-answerable questions.

3. Adaptation to abstention (IDK FT). IDK-15 adapter training with the same answerable questions and 15 percent canonical abstention examples. This is the proposed configuration.

The comparison between the Base Model and Normal FT allows us to see the effects of domain adaptation, whereas the comparison of Normal FT and IDK FT lets us see the effect of the abstaining data on calibration and hallucination because these two adapters have been trained under the same conditions.

The evaluation of all three models is performed on the IDK-15 test partition, which includes 449 queries: 370 are answerable and 79 are unanswerable. The reason why the IDK-15 test partition is used for all the models is that there are no unanswerable queries in the IDK-0 test partition and hence it cannot be used to evaluate the hallucination of unanswerable queries. Since the two mixtures were mixed independently, 15 out of 449 test queries, which are answerable, are also found in the IDK-0 training partition.

5.2 Hallucination Metrics (Hallucination Rate, IDK Precision/Recall/F1, Token F1)

For every test query, we check whether the model refused to answer, using a pattern of refusal phrases such as *"I don't have"*, *"I cannot"*, and *"don't have enough information"*. Combined with whether the query is answerable, this gives four outcomes:

	Model abstains	**Model answers**
Unanswerable	True Positive (TP)	Missed abstention (FN)
Answerable	False refusal (FP)	Answer attempt (TN)

Any answer to an unanswerable query is counted as a hallucination, since the source material contains no correct answer. The hallucination rate (HR) is the share of unanswerable queries that the model answers:

$$\mathrm{HR} = \frac{\mathrm{FN}}{\mathrm{TP} + \mathrm{FN}}. \tag{10}$$

IDK precision, recall, and F1 treat refusal as a binary classification task:

$$P_{\mathrm{IDK}} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \tag{11}$$

$$R_{\mathrm{IDK}} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \tag{12}$$

$$F1_{\mathrm{IDK}} = \frac{2\,P_{\mathrm{IDK}}\,R_{\mathrm{IDK}}}{P_{\mathrm{IDK}} + R_{\mathrm{IDK}}}. \tag{13}$$

Precision decreases when the model rejects requests it is supposed to respond to, and recall decreases when the model responds to requests it is supposed to reject. HR equals $1 - R_{\text{IDK}}$, however, it is reported separately since it shows the probability of receiving a hallucinated response directly. F1 is selected as the metric since the model receives full recall if it always rejects and no false rejections if it never rejects. False rejections, missed abstentions, and rejection ratio are also reported.

However, reducing hallucinations is only meaningful if the model responds to the answerable questions in a proper way. Therefore, answer quality is evaluated by token-level F1 between generated answer and reference, calculated as in SQuAD [40]. The texts are put into lowercase form, punctuation is stripped, and F1 is computed over overlapping tokens. F1 is selected instead of exact match since the model generates the response in free form, thus even the right answer in different phrasing will receive zero under exact match. Exact match is also reported. Answerable accuracy is additionally reported, which is the share of generated answers covering more than 40% of the reference.

5.3 Calibration Metrics (ECE, Brier, AUROC)

The confidence score for each response is the geometric mean of its per-token probabilities under greedy decoding [31]:

$$c(q) = \exp\left(\frac{1}{T}\sum_{t=1}^{T}\log p_\theta\left(r_t \mid r_{<t}, q\right)\right). \tag{14}$$

Every confidence score comes with a binary correctness label $y_i \in \{0, 1\}$, which considers the refusals. In the case of an unanswerable question, the response is considered to be correct if there is a refusal. When it comes to an answerable question, a refusal is treated as incorrectness, and the answer will be correct if the token F1 between the answer and the reference exceeds 0.5. According to this definition, a model that refuses all questions will be correct on 17.6% of test questions, which are unanswerable.

Expected Calibration Error (ECE) [14,34] splits the confidence range $[0, 1]$ into $M = 10$ equal-width bins and averages the gap between mean confidence and accuracy in each bin, weighted by bin size:

$$\text{ECE} = \sum_{m=1}^{M}\frac{|B_m|}{N}\left|\text{acc}(B_m) - \text{conf}(B_m)\right|. \tag{15}$$

Reliability diagrams are also plotted, since ECE does not show whether a model is over-confident or under-confident.

Brier score [4] is the mean squared error between confidence and correctness:

$$\text{Brier} = \frac{1}{N}\sum_{i=1}^{N}\left(c_i - y_i\right)^2. \tag{16}$$

It is not contingent on how the bins have been chosen.

AUROC is the area under the ROC curve when the confidence level is used for separating true responses from incorrect ones. This area corresponds to the probability of

the case when a randomly selected true response will have greater confidence level than a randomly selected false response, meaning that 0.5 would mean that the confidence level contains no information. While a model could be well-calibrated, but have equal confidence levels for true and false responses, AUROC will show if it can be utilized for spotting hallucinations.

5.4 Implementation Details

The pipeline is implemented using Python and PyTorch [36] and Hugging Face frameworks: transformers [44] for loading the model and generation, peft [33] for LoRA adapters, trl [43] (SFTTrainer) for SFT, bitsandbytes [7,8] for 4-bit quantization and 8-bit paged optimization, and datasets for dataset handling. AUROC scores are estimated using scikit-learn [37] framework, while ECE, Brier score, token-level F1, and the abstention confusion matrix are counted with custom scripts.

Training and evaluation were performed using only one NVIDIA RTX 3050 Ti laptop GPU with 4 GB VRAM. Because two 4-bit versions of Phi-3-mini model do not fit in VRAM at once, evaluation is performed sequentially. It means that the model is loaded, evaluated over the entire test dataset, and then unloaded, and the CUDA cache is cleaned up before loading another model. Adapters are saved separately from the frozen base model and attached to the model on loading, which ensures the same quantized base model parameters for all systems.

During the inference stage, both the models get the query in the chat format from the training data

```
<|user|>
{question}<|end|>
<|assistant|>
```

and are decoded using the greedy decoder, without sampling in order to make all predictions deterministic. The hallucination experiment (Section 5.2) sets the maximum limit of tokens to 100 so that no long answer is truncated during scoring. The calibration experiment (Section 5.3) on the other hand, sets the token limit to 64 and keeps the per-token log-probability to calculate the confidence score. Patterns of refusals used to label the output from the models are identical to the refusal patterns used to label the reference answers; in the case of IDK-15 test set, the exact same 79 unanswerable questions were identified by the refusal patterns during the collection of the data.

All the random functions like data shuffling, sampling of abstentions, split into train/valid/test, and hold-out valid use the seed of 42. With the two files provided above, the scripts will be able to replicate the partitions in Table 1.

6 Results and Analysis

This part presents the impact of each configuration on the two main aims of the research, i.e., hallucination reduction (Section 6.1) and confidence calibration (Sections 6.2 and 6.3), along with the impact analysis of the cost of such solutions in terms of false refusals (Section 6.4) and ablation of the contribution of the abstention data (Section 6.5). All

findings presented below are provided for the IDK-15 split, which contains 449 questions, of which 370 are answerable and 79 are unanswerable. The refusal metrics are computed using 100-token hallucinations analysis, and the calibration metrics are computed using 64-token calibration analysis (Section 5.4).

6.1 Hallucination Reduction

Table 2 gives the refusal characteristics for each system, while Figure 1 shows them visually.

Table 2. Hallucinations and refusals on the IDK-15 test set. HR refers to the rate of hallucinations (the number of missed abstentions divided by the 79 non-answerable questions). The answerable accuracy was measured according to the >40% coverage threshold (Section 5.2).

System	Refusal rate	IDK precision	IDK recall	IDK F1	HR	Missed abstentions (FN)	False refusals (FP)	Answerable acc.
Base	0.345	0.252	0.494	0.333	0.506	40	116	0.227
Normal FT	0.000	0.000	0.000	0.000	1.000	79	0	0.451
IDK FT	0.185	0.819	0.861	0.840	0.139	11	15	0.408

The most interesting result is how fine-tuning impacts hallucination. Before fine-tuning, the base model rejects about half of unanswerable questions (HR = 0.506). After fine-tuning on answerable domain-only dataset, it rejects none of them: the Normal FT model rejected none of the 449 test questions. The model answered all 79 of unanswerable questions, including those about individual student profiles, live schedules and non-existent courses, which by definition are all hallucinations. Domain adaptation improved domain knowledge of the model, increasing accuracy of answering on answerable questions from 0.227 to 0.451, while conditioning the model to respond to all queries, no matter if there is a correct answer to them or not. In terms of hallucinating in response to unanswerable questions, fine-tuning performed worse than no fine-tuning at all.

The inclusion of 15% of abstention results negates this phenomenon. Indeed, the IDK FT model abstains from answering 68 out of 79 unanswerable queries, which results in a decrease in the number of hallucinations to 0.139, a figure corresponding to a relative decline of 86% compared with Normal FT and 73% compared with the baseline model. Moreover, the model abstains in well-targeted cases, as 68 out of 83 refusals of the IDK FT model are made for the right reasons (precision 0.819), resulting in an IDK F1 score of 0.840. Overall, 18.5% of refusals of the model correspond to the real percentage of unanswerable queries in the test set (17.6%).

The recall of 0.494 shown by the base model is not to be construed as an indication of knowledge-awareness. Out of a total of 155 refusals given by it, 116 come in response to answerable queries; hence, almost 75% of its refusals are inappropriate. Most of its refusals are nothing but generic statements regarding access to institutionalized information, which have been made without any concern about the answerability of the given query. Figure 2 gives a comparison of the error patterns of the three models, with the base model erring mainly due to being overly cautious, Normal FT solely because

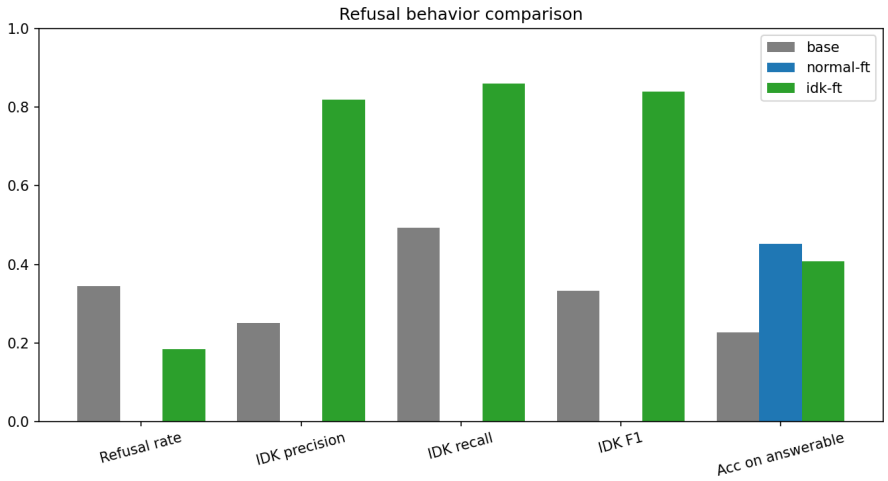


Fig. 1. Comparison of refusal behaviour (refusal rate, IDK precision, recall, F1, and answerable accuracy) for the base model, Normal FT, and IDK FT.

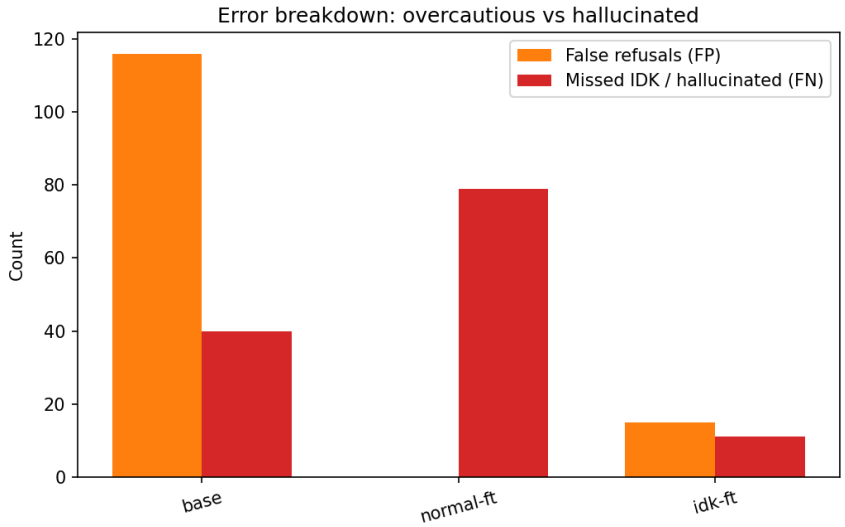


Fig. 2. Error breakdown showing false refusals (over-caution) and missed abstentions (hallucinations) for each system.

of hallucination, and IDK FT making a few errors of both kinds.

6.2 Calibration Improvement

The calibration statistics along with the quality-of-answer ratings that were derived from the test are presented in Table 3.

Table 3. Calibration and answer quality on the IDK-15 test set, using refusal-aware correctness with a token-F1 threshold of 0.5.

Metric	Base	Normal FT	IDK FT
Accuracy (refusal-aware)	0.100	0.327	0.477
Exact match	0.000	0.051	0.189
Token F1	0.168	0.391	0.515
ECE	0.622	0.408	0.304
Brier score	0.480	0.351	0.285
Reference NLL	2.405	1.663	0.965
AUROC	0.557	0.764	0.810

The calibration gets better at every step. ECE drops from 0.622 for the baseline model to 0.408 for Normal FT and then to 0.304 for IDK FT, a total decrease of 51%, out of which 25% is due to the addition of abstentions. The Brier score follows the same pattern ($0.480 \rightarrow 0.351 \rightarrow 0.285$), indicating that this is not just an artifact of binning.

The reliability plot in Figure 3 helps to understand the reason for these improvements. In fact, all three curves lie below the diagonal throughout the whole confidence range, which means that all three systems suffer from over-confidence and there is no way to remove the over-confidence completely. However, the form of the curves is quite different. The curve of the basic system is nearly flat, since the accuracy of this system varies only from 0 to 0.15 regardless of whether the confidence of the system equals 0.57 or 0.93; besides, the highest-confidence bin of this curve is zero, so the confidence of the system does not carry much information. The Normal FT curve rises with the increase in confidence but levels off at an accuracy of around 0.55 for confidence higher than 0.8, thus making the model wrong in nearly half of the most confident predictions made by it. The IDK FT curve rises monotonically over the entire interval, achieves an accuracy of 0.80 for a mean confidence of approximately 0.97, and thus comes closest to the diagonal out of the three models. There is still, however, a large difference in the middle range; at a confidence of approximately 0.65, IDK FT is correct in just 19% of cases.

There are two caveats that need to be taken into account while reading these results. First, some of the overall gains in performance are attributable to the unanswerable questions. Since the IDK FT repeats the canonical abstention exactly, any correct abstention gets a near-perfect token F1 and is generated with high confidence. Most of the gain in exact matches (from 23 out of 449 to 85 out of 449 questions) can thus be attributed to unanswerable questions, which is also reflected in the decrease in reference NLL, since all 79 unanswerable references use the same string that the model knows how to output. Second, the increase in refusal-aware accuracy (from 0.327 to 0.477, or about 67 additional accurate responses) roughly equals the number of unanswerable questions answered correctly. Given that IDK FT answers unanswerable questions in

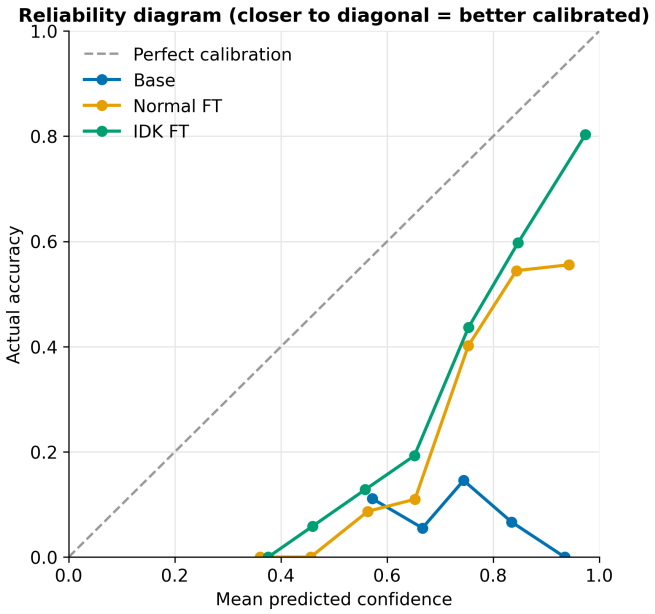


Fig. 3. Reliability diagram for the base model, Normal FT, and IDK FT (10 equal-width bins). The dashed diagonal denotes perfect calibration.

the same manner as in the hallucination detection, its accuracy on answerable questions with respect to F1 is close to 0.39, which is essentially the same as Normal FT's 0.40.

6.3 Confidence Separation (Histograms, AUROC)

While calibration tests the correctness of confidence on average, in addition to this, for deployment confidence must distinguish between correct and incorrect answers well enough to have a useful threshold. Figure 4 illustrates the distribution of confidence for correct and incorrect answers, while Figure 5 presents the ROC curves.

For the base model, the distributions are nearly identical. Both correct and incorrect answers lie in the same range between about 0.55 and 0.85, and the few correct answers (about 45) cannot be distinguished from the many more incorrect answers (about 400). There is no way to separate the two classes, as illustrated by an AUROC of 0.557, slightly above chance.

The fine-tuning process differentiates the two distributions. In normal FT, the correctly generated answers move more toward the 0.7–1.0 bin, while incorrectly generated answers are mostly in the 0.55–0.75 bin. However, the incorrect distribution still has an extended tail up to 0.95, which includes confident incorrect answers including the answers generated by the model for unanswerable questions. Its AUROC score is 0.764.

IDK FT performs best (AUROC 0.810). Its erroneous answers are located at the level of 0.6 confidence, while its correct answers are mostly in the range above 0.7, with an evident peak of over 80 correct answers in the 0.95–1.0 range, including mostly

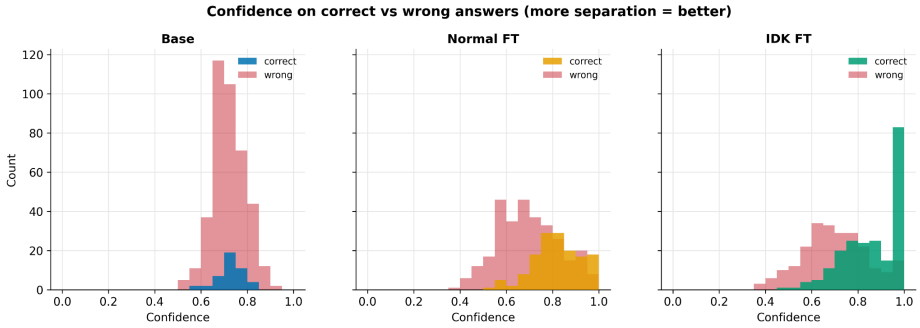


Fig. 4. Confidence distributions for correct and incorrect responses for each system.

canonical abstains. This peak has a practical importance. The ROC curve of IDK FT almost reaches 0.28 true positive rate vertically from the origin with almost zero false-positive rate. It means that about a quarter of correct answers of IDK FT may be used at the high confidence level without accepting even one incorrect answer in comparison with 0.05 for Normal FT. This tendency is observed in the whole low false-positive area: at the false-positive rate of 0.1, IDK FT keeps approximately 43% of its correct answers, while Normal FT only 25%. At a false-positive rate of about 0.4 and above, both curves converge. The gain is obtained only in the high-confidence region, which corresponds exactly to a practical threshold level.

However, the peak also has a few false positives (around 15 in the 0.95 to 1.0 bucket), and this number corresponds with the number of false refusals of 15 mentioned in Table 2. It is in line with the issue highlighted in Section 4.4 because the canonical expression for refusal is static in nature, and thus, the system can generate such an expression with high confidence even when refusal does not apply.

6.4 Cost of Abstention: False Refusals

Training a model to avoid makes a corresponding risk that the model will also reject queries that it can handle. This risk is measured directly.

The IDK FT rejects 15 of 370 answerable queries, with an FRR of 4.1%, which is much lower compared to that of the base model (FRR = 31.4%; 116 rejected queries) but just slightly higher than that of the Normal FT model (FRR = 0%). The coverage criterion indicates a decrease in answerable accuracy by 0.451 – 0.408 = 4.3% (decrease in number of correct answers = 16; 167 → 151).

This is significant in terms of the number of reductions because 16 correct answers lost are nearly the same as the number of false refusals. If we look at the number of correctly answered questions after excluding the non-answerable queries by each model, we see that IDK FT was correct for 151 out of 355 (42.5%), and Normal FT 167 out of 370 (45.1%). Given the close value of the two, we can conclude that the penalty was paid because of the refusal of some answerable questions, rather than poor quality answers.

Overall, the tradeoff is decidedly in favor of abstention training. For 16 less accurate responses, IDK FT saves itself from making 68 false ones – avoiding roughly four

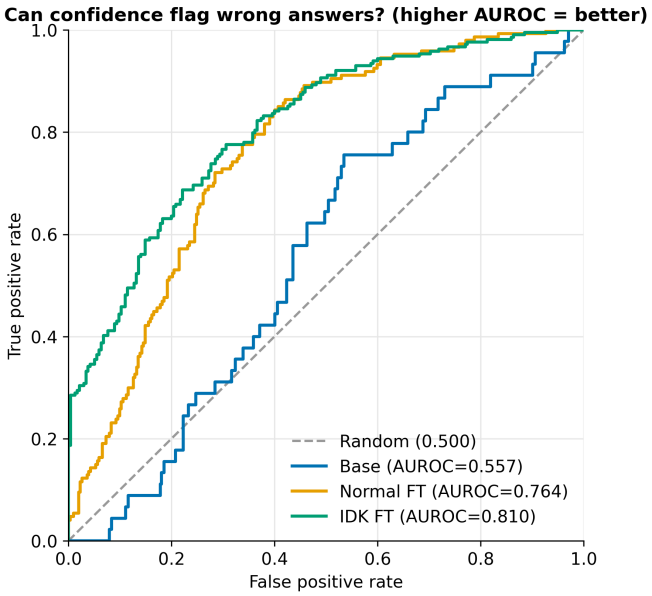


Fig. 5. ROC curves for discriminating correct from incorrect responses using sequence-level confidence.

hallucinations per answer sacrificed. In a student-oriented assistant, a false answer about exam dates, course credits, and registration can be quite damaging, but an unnecessary abstention will merely send a student to VTOP or their proctor.

One factor marginally favours Normal FT in this comparison. As noted in Section 5.1, 15 of the answerable test queries also appear in the training data of Normal FT but not in that of IDK FT. Normal FT's answerable accuracy may therefore be slightly overstated, and the true difference between the two systems may be smaller than 4.3 percentage points.

6.5 Contribution of Each Training Stage

As such, because Normal FT and IDK FT have identical base model, answerable data, hyperparameters, and random seed, comparing the two will highlight only the contribution of the abstention data. In the same way, comparing the base model to Normal FT highlights only the contribution of domain adaptation. These are shown in Table 4.

Table 4. Changes attributable to each stage.

Metric	Domain adaptation (Base → Normal FT)	Abstention data (Normal FT → IDK FT)
Hallucination rate	**+0.494 (0.506 → 1.000)**	**−0.861 (1.000 → 0.139)**
IDK F1	**−0.333**	**+0.840**
False refusal rate	**-0.314**	**+0.041**
Answerable accuracy	**+0.224**	**-0.043**
Token F1	**+0.223**	**+0.124**
ECE	**-0.214**	**-0.104**
Brier	**-0.129**	**-0.066**
AUROC	**+0.207**	**+0.046**

Both stages contribute in different ways. Domain adaptation makes the major contribution to the increase in domain knowledge: it almost doubles answerable accuracy, removes all refusals made by the base model randomly, and provides the highest individual improvements in ECE, Brier Score, and AUROC. On its own, it, however, removes abstention completely and leads to hallucinations on all unanswerable questions. The abstention data helps precisely in addressing this drawback and brings the hallucination ratio down from 1.000 to 0.139; it makes additional improvements in calibration and discrimination above those gained through domain adaptation, but only at the cost of 4.1% of false refusals and 4.3% of answerable accuracy.

Two aspects that should be pointed out as the limitations of this ablation are the following. First, the optimal abstention ratio could not be proven since only one ratio was tested, and thus, a smaller one could decrease false rejections even more without losing much hallucination reduction, while a larger ratio could result in an increased IDK recall but a larger number of false rejections. The second limitation is that the same random seed was used to train each setup.

7 Discussion

7.1 Interpretation

However, the key finding resulting from our experiment is that abstention behavior cannot result from FT alone. While standard FT increased answerable accuracy from 0.227 to 0.451, it did not refuse a single one out of 449 test prompts and provided answers for all 79 unanswerable questions. The baseline model refused approximately half of those questions (HR = 0.506), meaning that FT on domain-specific data even worsened the model's performance regarding hallucinations. The model learned that every prompt in the current domain had an answer and applied this behavior to the questions without any answers.

It took only the inclusion of refusals for the problem to be solved. The HR dropped from 1.000 to 0.139, with the rejection rate (18.5%) being close to the actual proportion of unanswerable questions in the testing dataset (17.6%). This means that the machine not only knows when there is no enough information but also knows how often it does so.

The analysis of calibration must be done carefully. ECE, Brier score and AUROC all improved at each step, however, as it is seen in Section 6.2, much of the change in the case of IDK FT is because of transforming confident mistakes into confident refusals to respond to unanswerable questions. On the questions that can be answered, IDK FT and Normal FT behave approximately the same way (0.39 and 0.40 according to the F1 criterion). Thus, abstention training affected how the model behaved when faced with unanswerable questions but did not affect the quality of the information it provided or the accuracy of its confidence. This is valuable nevertheless since the confident invented answer is the most dangerous flaw of such an assistant.

All three models remain stubbornly over-confident as well. At a confidence of about 0.65, IDK FT is correct only 19% of the time. The high-confidence peak covers the 15 false refusals. As the refusal is an inflexible sentence, it is possible to generate it with high probability, although it is not right to do so, and high confidence is no guarantee to find such errors.

The trade-off is clearly on the side of the IDK FT. It gives up 16 true predictions (167 to 151), but avoids 68 hallucinations, which means roughly four hallucinations avoided per every correct prediction given up. The reduction occurs mainly due to the 15 false refusals rather than poor quality of other answers. An incorrect answer about exam dates, credits, or registration might mislead the student, whereas a superfluous refusal will only make them contact VTOP or their proctor.

7.2 Qualitative Error Analysis

The errors of IDK FT fall into three categories.

The first category comprises the 11 false negatives, where the algorithm responded to a question it should have rejected. These questions mostly resemble typical answerable questions. Some mention non-existent but seemingly legitimate course codes and provide plausible titles and credit values for those. Others pertain to a detail not mentioned in any of the documents, e.g., a detail about a certain elective. These mistakes are the worst kind, as the answer sounds natural and is indistinguishable from the correct answer in any way. Their confidence scores may be reported, as a medium value could signal a possible cut-off threshold.

The second category of refusals is comprised of 15 incorrect refusals among 370 answerable questions. A possible explanation is that these questions include words which occur in the rejection dataset, such as “my”, “attendance”, “schedule” or “today”, or refer to programs or courses which the model was rarely exposed to. In such a case the model reacts to the question formulation rather than its content. As stated in Section 6.3, these refusals are produced with confidence more than 0.95, so they lie in the same area as correct refusals.

Third category: wrong answers to answerable questions. These constitute the biggest portion out of all three categories. Of the total 355 answerable questions attempted by IDK FT, 151 (42.5%) are answered correctly. The most common mistakes are incorrect numbers (credit values, course codes, etc.) and questions repeated verbatim without providing any factual information. Neither abstaining training has any effect on these errors, as they result from limited memorization capabilities of a 3.8B model on around 1,800 training examples.

One final test that can be made using these examples is to look at how the unanswerable questions were categorized (personal, in-the-moment, non-existent, and outside-the-scope). The model may simply be recognizing the types of questions being asked and not realizing what it does not know. If the abstentions that were missed come from one category and the refusals are correct in the other categories, then this would confirm this theory. The recall rate per category of refusal could tell us this and the data set has the information needed.

7.3 Limitations

There are a number of limitations to this study. First of all, only a single model (Phi-3-mini), only a single configuration per model, only a single seed per configuration and only a single ratio (15%) were tested. As a result, we cannot claim that 15% is the optimal ratio and we do not know how sensitive the results are to different runs. Another limitation concerns the size of our test set, which is rather small. HR = 0.139 is associated with 11 out of 79 questions without an answer, which results in a 95% interval of around 0.08 to 0.23; thus, differences of a few questions cannot be overinterpreted. Finally, 15 answerable questions in the test set are present in Normal FT training set.

Since refusals are recognized by a list of phrases, a refusal phrased differently will also be treated as a response. The model was deliberately set up conservatively, but this will impact all metrics where refusals play any role. All training refusals, in fact, use a single sentence. It is easily measurable, although the model might learn this as a script and not a means of expressing uncertainty, making it more confident about refusals than it should. Different wording would result in a different score.

Correctness is evaluated using token F1 (threshold at 0.5) and the 40% word-coverage criteria. These do not check whether the numbers in an answer (such as credit values or course codes) are correct, nor whether a phrase has been phrased differently, meaning the former would be accepted while the latter would be rejected. Accuracy scores should thus be used for comparisons and not as absolute numbers. An additional check for course codes and credit values would be more appropriate.

The data was collected from one institution, and the unanswerable questions were designed for specific categories instead of being collected from actual students. Since the split is random and not categorized by courses, the test questions are likely to be very similar to the training ones, and we do not measure how many. The findings might not be generalizable to other institutions or actual student queries.

Lastly, the confidence score is just the mean token probability, and no form of calibration, like temperature scaling, is applied; thus, the systems remain over-confident. The model also generates based on the learned knowledge without using retrieval. Using retrieval from the curriculum documents will most likely increase the accuracy of the answerable questions much more than the abstention training alone. Combining the two is a logical next step to take. The experiments were conducted on a 4 GB laptop GPU with 4-bit quantization, which put some constraints on model size and configuration.

8 Conclusion and Future Work

In this work, we have investigated the possibility of enhancing the confidence calibration of a small language model, as well as lowering its hallucination rate, through the construction of the fine-tuning data only. Phi-3-mini was fine-tuned using QLoRA for institutional questions under the constraint of 4 GB of VRAM, and the comparison is made between three models, namely the base model, a standard domain fine-tuned model, and an abstention-aware fine-tuned model in which 15% of the training data comprised abstentions with one canonical response only.

There are three main conclusions. Firstly, domain fine-tuning alone is not enough, and when used on unanswerable queries causes hallucinations. When applied to answerable queries, domain fine-tuning approximately doubled accuracy on answerable queries, but made the model not capable of abstaining; standard fine-tuning generated answers to all 79 unanswerable queries in the test set, thus raising the hallucination rate from 0.506 in the baseline setting to 1.000. Secondly, controlled inclusion of the abstention data considerably decreases the hallucination rate. In the case of 15% abstention examples in the dataset, the hallucination rate went down to 0.139 and IDK F1 score became 0.840. The price of such an improvement was quite small: there were 15 false refusals for 370 answerable queries (4.1%), and 4.3 percentage point decrease in accuracy on answerable queries – mainly due to refusals, and not due to quality of answers. Thirdly, the same modification of the training data improves calibration and discrimination without using any calibration objective. ECE decreased from 0.622 (baseline) to 0.408 (domain fine-tuning) and 0.304 (abstention-aware domain fine-tuning), while AUROC increased from 0.557 to 0.764 and 0.810.

There is a recognition of the limitations of these findings in the analysis. All models retain their over-confidence throughout the whole range of confidence levels. Some of the improvements in the calibration process are based on the model's ability to repeat a particular fixed phrase as abstention with a very high confidence level, and in the same way, some false abstentions are assigned a very high level of confidence. These results are generated using just one abstention rate, one seed value, one base model and one institution's data, and evaluated through lexical overlap.

The problems described point towards some possible lines of future research:

- *Abstention ratio calibration.* Through training models at different abstention ratios (such as 5-30%) with different random initialisations, it will be possible to see how calibration, hallucination rates, and false refusals differ with respect to the ratio of the abstention samples and whether 15% is close to the optimal value.
- *Context dependent varying vs canonical abstentions.* Training the model on context-dependent varying abstentions (for example, asking the user to go to VTOP or the proctor) will determine whether it is possible to teach the model the idea of abstention as an uncertainty indicator instead of rote memorisation of certain phrases and its impact on the refusal confidence level.
- *Post-hoc calibration and selective prediction.* The fit of temperature scaling on the validation set will reduce over-confidence. Then, the confidence threshold will need to be determined.
- *Additional uncertainty metrics.* Comparison between confidence levels (probabil-

ity of token prediction) and sampling-based semantic alignment, verbalizing confidence, and probing of latent space can give some clue about the necessity of the latter when using limited capacity models.
- *Measuring the correctness.* Usage of NLI evaluation and LLMs as evaluators, and human evaluations, if needed on top of that, will allow researchers to avoid token overlap, which can give an idea about the percentage of "wrong" answers that are in fact correct paraphrases.
- *Combination with the retrieval mechanism.* Combining the abstain-capable model with the documents from the curriculum will allow answering all answerable questions while abstaining from the rest.
- *Generalization.* Conducting the same experiment with other small models (e.g. sub-4B Llama, Qwen, Gemma, and others) and documents from other educational institutions will allow making sure about the generalization of the findings.

In summary, the experiments showed that by making a small modification to the training data, a small and cheap model could become noticeably more calibrated and significantly less likely to generate errors in hallucination for the specific domain. Where the use of the model may affect students, as an error could have consequences in regard to the number of credits or deadline dates, being able to detect when the model does not know an answer is equally important to answering questions correctly.

Acknowledgments. We thank Swathi JN, School of Computer Science and Engineering, Vellore Institute of Technology, for guidance throughout this project.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Abdin, M., et al.: Phi-3 technical report: A highly capable language model locally on your phone. arXiv preprint arXiv:2404.14219 (2024)
2. Biderman, D., et al.: LoRA learns less and forgets less. Transactions on Machine Learning Research (2024)
3. Brahman, F., et al.: The art of saying no: Contextual noncompliance in language models. In: Advances in Neural Information Processing Systems (Datasets and Benchmarks Track) (2024)
4. Brier, G.W.: Verification of forecasts expressed in terms of probability. Monthly Weather Review 78(1), 1–3 (1950)
5. Cheng, Q., Sun, T., Liu, X., Zhang, W., Yin, Z., Li, S., Li, L., He, Z., Chen, K., Qiu, X.: Can AI assistants know what they don't know? In: Proceedings of the 41st International Conference on Machine Learning (ICML) (2024)
6. Chow, C.K.: On optimum recognition error and reject tradeoff. IEEE Transactions on Information Theory 16(1), 41–46 (1970)
7. Dettmers, T., Lewis, M., Shleifer, S., Zettlemoyer, L.: 8-bit optimizers via block-wise quantization. In: International Conference on Learning Representations (ICLR) (2022)
8. Dettmers, T., Pagnoni, A., Holtzman, A., Zettlemoyer, L.: QLoRA: Efficient finetuning of quantized LLMs. In: Advances in Neural Information Processing Systems (NeurIPS) (2023)
9. El-Yaniv, R., Wiener, Y.: On the foundations of noise-free selective classification. Journal of Machine Learning Research 11, 1605–1641 (2010)

10. Feng, S., Shi, W., Wang, Y., Ding, W., Balachandran, V., Tsvetkov, Y.: Don't hallucinate, abstain: Identifying LLM knowledge gaps via multi-LLM collaboration. In: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL) (2024)
11. Geifman, Y., El-Yaniv, R.: Selective classification for deep neural networks. In: Advances in Neural Information Processing Systems (NeurIPS) (2017)
12. Gekhman, Z., Yona, G., Aharoni, R., Eyal, M., Feder, A., Reichart, R., Herzig, J.: Does fine-tuning LLMs on new knowledge encourage hallucinations? In: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP) (2024)
13. Gunasekar, S., et al.: Textbooks are all you need. arXiv preprint arXiv:2306.11644 (2023)
14. Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q.: On calibration of modern neural networks. In: Proceedings of the 34th International Conference on Machine Learning (ICML). PMLR, vol. 70, pp. 1321–1330 (2017)
15. Houlsby, N., Giurgiu, A., Jastrzebski, S., Morrone, B., de Laroussilhe, Q., Gesmundo, A., Attariyan, M., Gelly, S.: Parameter-efficient transfer learning for NLP. In: Proceedings of the 36th International Conference on Machine Learning (ICML). PMLR, vol. 97, pp. 2790–2799 (2019)
16. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W.: LoRA: Low-rank adaptation of large language models. In: International Conference on Learning Representations (ICLR) (2022)
17. Hu, X., Zhang, Y., Wei, S., Zhao, C., Li, Q., Li, B., Chen, F.: Small updates, big doubts: Does parameter-efficient fine-tuning enhance hallucination detection? arXiv preprint arXiv:2602.11166 (2026)
18. Huang, L., et al.: A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. ACM Transactions on Information Systems 43(2) (2025)
19. Ji, Z., et al.: Survey of hallucination in natural language generation. ACM Computing Surveys 55(12), 1–38 (2023)
20. Jiang, Z., Araki, J., Ding, H., Neubig, G.: How can we know when language models know? On the calibration of language models for question answering. Transactions of the Association for Computational Linguistics 9, 962–977 (2021)
21. Kadavath, S., et al.: Language models (mostly) know what they know. arXiv preprint arXiv:2207.05221 (2022)
22. Kalai, A.T., Nachum, O., Vempala, S.S., Zhang, E.: Why language models hallucinate. arXiv preprint arXiv:2509.04664 (2025)
23. Kalai, A.T., Vempala, S.S.: Calibrated language models must hallucinate. In: Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC) (2024)
24. Kamath, A., Jia, R., Liang, P.: Selective question answering under domain shift. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL) (2020)
25. Kapoor, S., et al.: Large language models must be taught to know what they don't know. In: Advances in Neural Information Processing Systems (NeurIPS) (2024)
26. Kuhn, L., Gal, Y., Farquhar, S.: Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In: International Conference on Learning Representations (ICLR) (2023)
27. Lester, B., Al-Rfou, R., Constant, N.: The power of scale for parameter-efficient prompt tuning. In: Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP) (2021)
28. Lewis, P., et al.: Retrieval-augmented generation for knowledge-intensive NLP tasks. In: Advances in Neural Information Processing Systems (NeurIPS) (2020)
29. Li, X.L., Liang, P.: Prefix-tuning: Optimizing continuous prompts for generation. In: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics

(ACL), pp. 4582–4597 (2021)
30. Lin, S., Hilton, J., Evans, O.: TruthfulQA: Measuring how models mimic human falsehoods. In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL), pp. 3214–3252 (2022)
31. Malinin, A., Gales, M.: Uncertainty estimation in autoregressive structured prediction. In: International Conference on Learning Representations (ICLR) (2021)
32. Manakul, P., Liusie, A., Gales, M.J.F.: SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP) (2023)
33. Mangrulkar, S., Gugger, S., Debut, L., Belkada, Y., Paul, S., Bossan, B.: PEFT: State-of-the-art parameter-efficient fine-tuning methods. `https://github.com/huggingface/peft` (2022)
34. Naeini, M.P., Cooper, G.F., Hauskrecht, M.: Obtaining well calibrated probabilities using Bayesian binning. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 29 (2015)
35. OpenAI: GPT-4 technical report. arXiv preprint arXiv:2303.08774 (2023)
36. Paszke, A., et al.: PyTorch: An imperative style, high-performance deep learning library. In: Advances in Neural Information Processing Systems (NeurIPS) (2019)
37. Pedregosa, F., et al.: Scikit-learn: Machine learning in Python. Journal of Machine Learning Research 12, 2825–2830 (2011)
38. Porwal, S., Prasath S, S., Bi, X., Scandlen, M.: Evaluating hallucinations in domain-adapted large language models. arXiv preprint arXiv:2606.07521 (2026)
39. Rajpurkar, P., Jia, R., Liang, P.: Know what you don't know: Unanswerable questions for SQuAD. In: Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL), pp. 784–789 (2018)
40. Rajpurkar, P., Zhang, J., Lopyrev, K., Liang, P.: SQuAD: 100,000+ questions for machine comprehension of text. In: Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. 2383–2392 (2016)
41. Tian, K., Mitchell, E., Zhou, A., Sharma, A., Rafailov, R., Yao, H., Finn, C., Manning, C.D.: Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP) (2023)
42. Wen, B., Yao, J., Feng, S., Xu, C., Tsvetkov, Y., Howe, B., Wang, L.L.: Know your limits: A survey of abstention in large language models. Transactions of the Association for Computational Linguistics (2025)
43. von Werra, L., et al.: TRL: Transformer reinforcement learning. `https://github.com/huggingface/trl` (2020)
44. Wolf, T., et al.: Transformers: State-of-the-art natural language processing. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, pp. 38–45 (2020)
45. Xiong, M., Hu, Z., Lu, X., Li, Y., Fu, J., He, J., Hooi, B.: Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs. In: International Conference on Learning Representations (ICLR) (2024)
46. Yang, Y., Chern, E., Qiu, X., Neubig, G., Liu, P.: Alignment for honesty. In: Advances in Neural Information Processing Systems (NeurIPS) (2024)
47. Zhang, C., Zhu, Z., Wei, Y., Tian, B., Liu, J., Wang, H., Wang, X., Liu, Y.: Confidence-calibrated small-large language model collaboration for cost-efficient reasoning. arXiv preprint arXiv:2603.03752 (2026)
48. Zhang, H., Diao, S., Lin, Y., Fung, Y.R., Lian, Q., Wang, X., Chen, Y., Ji, H., Zhang, T.: R-Tuning: Instructing large language models to say 'I don't know'. In: Proceedings of the 2024

Conference of the North American Chapter of the Association for Computational Linguistics (NAACL) (2024)